Wanshu Zhang

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Computational history • NLP & LLM evaluation • information networks • digital systems

Research profile. PhD researcher at the Luxembourg Centre for Contemporary and Digital History (C²DH), University of Luxembourg. I study how uncertain claims change through repetition, attribution, correction, and reuse, and I build reproducible computational workflows for noisy historical text. My work combines NLP and LLM evaluation, annotation and benchmark design, provenance/source-reuse networks, large-scale corpus processing, and public-facing digital systems.

AI / NLP
LLM annotation and evaluation; event/entity extraction; semantic similarity; OCR-aware processing.

Scale / infrastructure
Large historical corpora, streaming data, DuckDB, Hugging Face datasets, ULHPC/SLURM workflows.

Open research software
RumourFlow provenance/reuse toolkit; RecycloLux multilingual serious-game system.

Evaluation / evidence
Gold-set design, inter-annotator agreement, error analysis, source/reuse networks, reproducible validation.

EDUCATION

University of Luxembourg — Luxembourg Centre for Contemporary and Digital History (C²DH) Jul 2023–present
PhD researcher in History / Digital History
- Thesis: Repetition Patterns and Ephemeral Effects of Rumors (Collective Amnesia Phenomenon).
- Coursework includes Digital Epistemology and Narratives of the Digital.

Higher School of Economics (HSE), Moscow Sep 2020–Jun 2022
M.A. in Linguistic Theory and Language Description
- Thesis: Semantic Fields of Pulling and Pushing in Mandarin Chinese.
- Training included formal approaches to language, Python, quantitative analysis and visualization, sociolinguistics, typology, linguistic anthropology, and neurolinguistics.

Henan Institute of Science and Technology Sep 2014–Jun 2018
B.A. in Biological Engineering
- Thesis: Cloning and Expression Analysis of the TaLr67 Gene in Wheat.

PEER-REVIEWED PUBLICATIONS AND ACCEPTED CONTRIBUTIONS

Zhang, W. (2026). "Validating Rule-Based Classification of Historical Fake News with Event Extraction: A Case Study of the Great Moon Hoax of 1835." In X. Ding & Y. Dong (Eds.), Computational Linguistics and Natural Language Processing – 2nd International Conference, CLNLP 2025, Revised Selected Papers, 39–46. Springer Nature. DOI: 10.1007/978-981-95-4788-3_4

Zhang, W. (2026). "LLM-Based Frame and Stance Annotation for 19th-Century Rumour Discourse in US and UK Newspapers." Proceedings of NeoLLM 2026: Neology and Large Language Models. DOI: 10.63317/4qdnrzfmibbz

Zhang, W. (2026). "The Editor's Dilemma: A Browser-Based Serious Game Prototype for Teaching Epistemic Vocabulary Through Historical Misinformation." Joint Conference on Serious Games 2026. [accepted; forthcoming]

Zhang, W. (2026). "RecycloLux: A Multilingual Serious Game Demo for Civic Waste Sorting Literacy in Luxembourg." Joint Conference on Serious Games 2026, Lecture Notes in Computer Science proceedings. [accepted; forthcoming]

Zhang, W. (2026). "Linking Rumour Flows: A Gazetteer-Based Toolkit for Source and Reuse Detection in Historical Newspapers." EADH 2026 poster abstract. [presented] DOI: 10.5281/zenodo.23267368

Zhang, W., Liao, M., & Wang, Y. (2026). "The Anatomy of Digital Haunting: Extracting the Universal Mechanism of Fear from a Case Study of Northeast China's Dreamcore." Spatial Humanities 26 abstract. [presented]

SELECTED RESEARCH, SOFTWARE, AND DIGITAL SYSTEMS

PRESTO — discovering and tracking rumoured events in historical newspapers
- Designed a two-phase workflow for high-recall discovery followed by event-focused extraction and validation across large historical newspaper corpora.
- Integrated rule-based NLP, dependency/event features, OCR-aware preprocessing, embeddings, and LLM-assisted classification; presented at Digital Humanities 2026.

RumourFlow — source, reuse, and provenance detection in historical news
GitHub | EADH 2026 poster
- Developed a review-oriented toolkit combining source cues, cross-collection gazetteers, text-reuse detection, clustering, and graph-ready exports for historical newspaper research.
- Methodological focus: distinguishing dependent repetition from independent corroboration and keeping inferred connections auditable.

LLM frame and stance annotation for nineteenth-century rumour discourse
- Constructed US/UK newspaper datasets and annotation schemes for frames, stance, and anachronism; evaluated LLM-based labeling against manually checked data.
- Emphasis: reproducible evaluation, error analysis, cross-national comparison, and semantic change.

Helsinki Digital Humanities Hackathon — #Books May 2026
- Co-built a manually annotated 1,200-record VD16/VD17 dataset with a medieval-literature domain expert; compared GPT-5.5 and Gemini 3.1 Pro prompting strategies and developed a 3D prism visualization for semantic change.

RecycloLux — multilingual civic waste-sorting serious game 2026

Live system | GitHub

- Designed and deployed a source-verified multilingual browser game with 91 household items, age-differentiated pools, three difficulty levels, explanatory feedback, search/reporting workflows, and privacy-aware maintenance.
- Accepted as a JCSG 2026 demo paper.

Birthday Wishes Collector — lightweight team web utility 2026

GitHub

- Built an HTML/CSS/JavaScript browser tool for upcoming-birthday calculation, structured message collection, and Google Forms submission without OAuth.

SELECTED PRESENTATIONS AND ADVANCED TRAINING

Digital Humanities 2026 (DH2026), Daejeon, South Korea Jul 2026

- Presented PRESTO.

European Association for Digital Humanities Conference (EADH 2026), Kraków, Poland Sep 2026

- Presented poster “Linking Rumour Flows.”

Spatial Humanities 26, University of Minho, Braga, Portugal Sep 2026

- Presented “The Anatomy of Digital Haunting.”

Digital Humanities Summer Institute (DHSI), Université de Montréal Jun 2026

- Week 1: Introduction to Linked Open Data and the Semantic Web (8–12 Jun).
- Week 2: AVAnnotate Open Source Application for Audiovisual Digital Exhibits and Editions (15–19 Jun).

Erasmus+ Blended Intensive Programme — A Journey into Knowledge and Reasoning, Silesian University of Technology May–Jul 2026

- Training in logic, knowledge representation, constraint programming, rule-based transformation, automata, argumentation, and interdisciplinary project work; in-person component in Gliwice, 6–10 Jul 2026.

NeoLLM 2026, Palma de Mallorca May 2026

- Presented “LLM-Based Frame and Stance Annotation.”

European Summer University in Digital Humanities, Besançon, France Jul–Aug 2025

Deep Data Science for Digital History Summer School, University of Agder / Lesbos Jun 2025

EUROPAST Summer School — Facing the Past: Public History for a Stronger Europe Jul 2024

HONORS AND AWARDS

Best Oral Presentation — 2nd International Conference on Computational Linguistics and Natural Language Processing (CLNLP 2025) 2025

Full Scholarship — Higher School of Economics 2020

Chinese National Inspirational Scholarship 2016

Second Prize — Eighth Henan Translation Competition 2016

Second Prize — 8th National College Students Ocean Knowledge Competition 2015

METHODS AND TECHNICAL SKILLS

Area	Experience
NLP / LLMs	LLM-assisted annotation and evaluation; prompt-strategy comparison; stance/frame classification; event/entity extraction; semantic similarity; topic modeling; OCR-aware processing.
Tools	Python, R, pandas, spaCy, Stanza, Stanford CoreNLP, SentenceTransformers, scikit-learn/Gensim, Hugging Face datasets, DuckDB.
Infrastructure	ULHPC/SLURM; large streaming datasets; reproducible batch processing and checkpointing.
Evaluation	Annotation design, gold-set construction, inter-annotator agreement, error analysis, source/reuse networks, quantitative visualization.
Web / prototyping	HTML, CSS, JavaScript; progressive web apps; browser-based interactive tools; lightweight Apps Script / Google Forms integrations.
Languages	Chinese (native); English (fluent).